Imam Mohammad Ibn Saud Islamic University College of Computer and Information Sciences Information Systems Department **Course Project** **Couse ID:** **Course name:**	**Given Mark:** **20**			
	☐**Proposal**	☐**Report**	☐**Demonstration/ Presentation slides**	☐**Group** ☐**Individual**

Students Name(s) & ID	**Project Instructions**
1.Reema alafif 445005560 2.Njoud Alturaiki 445005055 3.Ghada Alajmi 444007555 4.	√ I declare that the proposed document is my own work and I own the copy right of it with no copy rights violation or plagiarism from other resources. √ Submission due on: **13/5/2026** √ Cheating may cause your total mark to be **Zero** grade for the whole project. √ Only 30% of similar content is accepted from SafeAssign report. √ -1 mark for each late day after due date, and the project will not be accepted after the 3rd day.

	Rubrics	Given	Out of
Marking criteria	**Prototype**		**5**
	Good Redesign Sketches (low-fidelity protypes at least 2)		2
	Good Prototype design (high-fidelity protype)		3
	Total		**5**
	Report		**10**
	User Identification		1
	Task Identification		1
	Proper Matching between the task steps and Seven Stages of Actions		2
	Two Gulfs for each application		1.5
	Good identifications of UX issues		1
	Demonstration of good analysis skills		1
	Demonstration of good designing skills		1
	References		0.5
	Appendix (at least A.2)		0.5
	Following the template		0.5
	Plagiarism?		***>30% Minus -?***
	Late submission?		***-1 for each day? (3 days limit)***
	Total		**10**
	Demonstration		**5**
	Knowledge		1
	Skills		1
	Team work		1
	Handling questions		2
	Total		**5**
	Total Project Mark		**20**

Project overview

In the mobile travel industry, user experience (UX) design is critical to helping users navigate search tasks and evaluate booking terms without anxiety. This project presents a comparative UX evaluation of AlMosafer, an application designed for the Middle Eastern market, and Expedia, a global travel booking application. Specifically, we analyze how each application manages a sensitive, multi-step user task: finding and booking a "Fully Refundable" hotel room.

During our initial analysis, we discovered a major conceptual and structural discrepancy between the two platforms. While both applications offer a "Free Cancellation" filter on their search results page, AlMosafer does not treat "Free Cancellation" as synonymous with a cash refund. On AlMosafer, a free cancellation may result in non-refundable wallet points or credit. Furthermore, to verify the actual refund terms, the user is forced to leave the main search feed, enter the hotel details page, and navigate nested policy links. In contrast, Expedia provides explicit "Fully Refundable" cash-back tagging directly on the main search results card, allowing instant evaluation without extra clicks.

To systematically analyze and resolve this UX breakdown, this study is structured into four sequential phases:

- **Phase 1: Task Identification and Seven Stages of Action:** We formally map the steps of our selected booking task against Don Norman's Seven Stages of Action to pinpoint exactly where cognitive breakdowns, and navigation delays occur.

- **Phase 2: User Identification and System Testing:** We build a target traveler persona and conduct usability testing with a volunteer participant. By observing their interaction with both apps, we document user hesitation and verify the cognitive gap identified on AlMosafer.

- **Phase 3: Redesign:** To address AlMosafer's information-design issues, we apply the design principles of Discoverability and Feedback. We present hand-drawn low-fidelity sketches that bring clear cash-refund indicators directly to the primary search results feed.

- **Phase 4: Enhanced Prototype:** We transform our low-level design concepts into high-level design prototype screens. This demonstrates how our design interventions close the "Gulf of Evaluation" and drastically reduce cognitive load for cautious travelers.

Phase 1: Task Identification and Application of Seven Stages of Action:

Task Definition: Book a "Fully Refundable" hotel room in Dubai for two people.

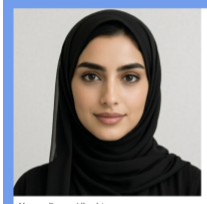
Stage of Action	AlMosafer Application	Expedia Application
1. Goal (form a goal)	The user wants to book a hotel in Dubai for their family that can be cancelled without financial penalty if plans change.	The user wants to book a hotel in Dubai for their family that can be cancelled without financial penalty if plans change.
2. Plan (the action)	The user decides to open the AlMosafer app, search for Dubai hotels, and then filter the results for free cancellation.	The user decides to open Expedia, search for hotels in Dubai, and then filter the results for refundable hotels.
3. Specify (an action sequence)	The user taps stays, enters destination, picks the dates, picks how many rooms and people, then taps search button and after the search the user filters the results.	The user taps stays, enters destination, picks the dates, picks how many rooms and people, then taps search button and after the search the user filters the results.
4. Perform (the action sequence)	The user inputs the basic search info, hits search, opens the filter sheet on the results page, checks "Free Cancellation," and taps apply.	The user inputs the basic search info, hits search, taps the filter bar on the results page, scrolls to locate the "Fully Refundable" filter, and selects it.
5. Perceive (the state of the world)	The user views the filtered hotel search results screen, looking for clear cash-refund indicators.	The user views the filtered search results screen, looking for refundability indicators.
6. Interpret (the perception)	The user sees "Free Cancellation" on the feed, but because "Free Cancellation" is not structurally identical to "Refundable" on this app, they are forced to tap on	The user sees a highly visible "Fully Refundable" tag directly on the results card with a clear green font, clarifying that it is a

	the hotel, scroll, and click to read the room policies to verify if they get their actual money back.	cash-back policy without requiring a page click.
7. Evaluate (the outcome with the goal)	The user feels anxious because they had to perform extra navigation steps to confirm if their financial risk is covered.	The user feels confident because the exact refund status was verified immediately on the primary screen.

Phase 2: User Identification and System Testing:

Presona pictures

Persona 1: The Cautious Student



Name: Reem Alharbi
Age: 24
Occupation: University Student
Location: Riyadh
Education: Bachelor's Student
Relationship status: Single
Personality Traits: Organized, careful, detail-oriented, anxious when booking online

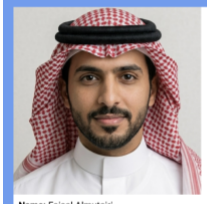
Goals:
1. Book hotels with clear and guaranteed refund policies.
2. Avoid losing money if academic or personal plans change.
3. Complete the entire booking process quickly via mobile.

Challenges:
- Gets confused by technical terms like "Free Cancellation" vs. "Fully Refundable."
- Finds long, text-heavy policy pages overwhelming and boring.
- Feels high stress when payment terms aren't explicitly clear.

Needs:
- Clear "Refundable" labels directly on the search results.
- A side-by-side comparison of different hotel options.
- A simplified booking flow with minimal steps.

Representative Quote:
"I just want to know if I'll get my money back without opening five different pages."

Persona 3: The Fast-Paced Professional



Name: Faisal Almutairi
Age: 31
Occupation: Sales Executive
Location: Riyadh
Education: Bachelor's Degree in Marketing
Relationship status: Single
Personality Traits: Busy, practical, fast decision-maker, tech-savvy

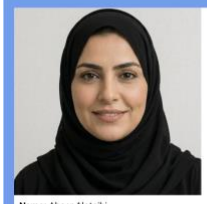
Goals:
1. Secure a hotel room instantly during high-pressure business trips.
2. Find refund policies at a glance without deep searching.
3. Use a reliable and "lag-free" travel application.

Challenges:
- Frustrated by "hidden" information or small print.
- No patience or time to read through lengthy terms and conditions.
- Wants immediate confirmation that the booking is successful.

Needs:
- Visible refund status displayed on the main dashboard.
- Advanced filtering system (e.g., "Sort by Refundability").
- A "One-Click" booking experience to save time.

Representative Quote:
"I need important booking information immediately; I don't have time to waste."

Persona 2: The Family Planner



Name: Abeer Alotaibi
Age: 48
Occupation: Mother & Part-Time Teacher
Location: Jeddah
Education: Bachelor's Degree in Education
Relationship status: Married
Personality Traits: Careful, responsible, patient, family-oriented

Goals:
1. Find comfortable, family-friendly hotels with specific amenities.
2. Maintain flexibility in case one of the children gets sick or plans shift.
3. Ensure no financial loss due to strict or tricky cancellation rules.

Challenges:
- Feels overwhelmed by complex and detailed hotel legal jargon.
- Experiences high anxiety right before hitting the "Pay" button.
- Finds it difficult to locate where to start a refund if needed.

Needs:
- Large, color-coded "Fully Refundable" badges.
- Transparent, step-by-step breakdown of the refund process.
- Human-centric design with easy-to-read fonts and icons

Representative Quote:
"I want to feel safe and 100% sure about my booking before I commit to paying."

Participants Sheet :

Participant ID	Age	Gender	Experience Level	Device	Persona Mapping	Occupation
P1	24	Female	Moderate	iPhone	The Cautious Student (Reem)	University Student
P2	31	Male	High	Android	The Fast-Paced Professional (Faisal)	Sales Executive
P3	48	Female	Moderate	iPhone	The Family Planner (Abeer)	Teacher

Test Task (Scenario):

Find a hotel in Dubai and ensure the booking is fully refundable, and proceed to the payment page without paying.

P1 (Age 24 – Student Type)

- **AlMosafer:**

Found the application easy to use overall, but had to open hotel pages or scroll further to confirm the refund details. The "Free Cancellation" information was visible, but the exact refund conditions were not immediately clear.

- **Expedia:**

Preferred this application because the "Fully Refundable" status appeared directly next to the hotel name on the search results page. She also found it helpful that nearby landmarks and attractions were displayed clearly.

- **Observation:**

This layout allowed her to make decisions quickly without opening the hotel details page. She relied on:

- Map visibility
- Nearby landmarks
- Immediate refund labels

P2 (Age 31 – Professional Type)

- **AlMosafer:**

Felt that the interface was slightly slower for his needs because refund information was not placed at the top of the listing. He needed additional steps to verify whether the hotel was fully refundable.

- **Expedia:**

Preferred the fast-paced interaction flow. Refundable hotels and "Book Now, Pay Later" options were clearly displayed beside the hotel name, reducing the need for deep reading.

- **Cognitive Gap:**

On AlMosafer, the participant almost experienced a user oversight because he expected refund information to appear immediately. and Since the information was less visible, he nearly selected a non-refundable option unintentionally.

P3 (Age 48 – Family Planner Type)

- **AlMosafer:**

Carefully read all details before making a decision. She focused first on finding hotels suitable for children and family needs. Although the familiar layout helped but she still had to search within the description to locate the refund policy. She noticed that some hotels only allowed free cancellation before a specific deadline, which increased hesitation and confusion.

- **Expedia:**

Even though she carefully reviewed the Terms & Conditions, seeing the "Fully Refundable" label directly beside the hotel name gave her an immediate sense of reassurance. She also appreciated that refundability and "Book Now, Pay Later" options were clearly visible before opening the hotel page.

- **The Review Factor:**

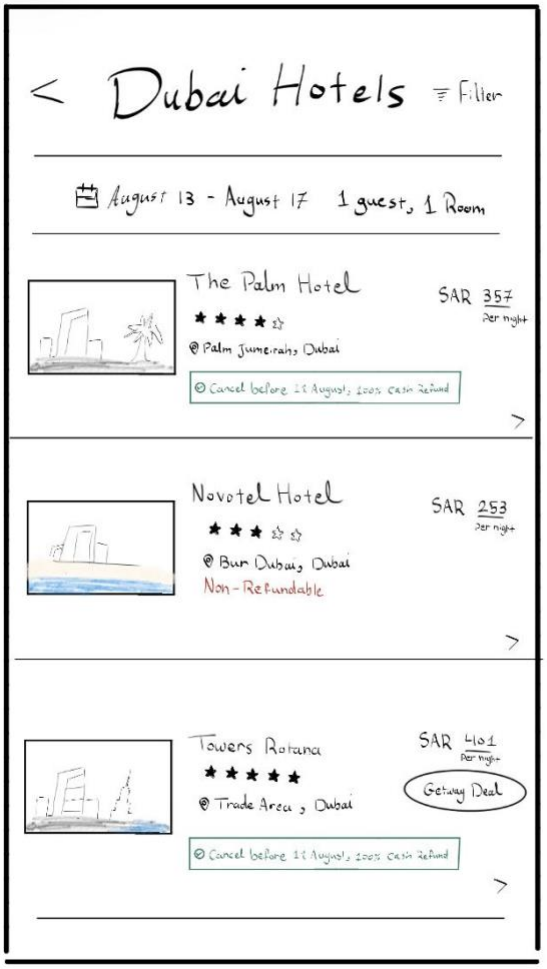
Both applications included reviews, but she relied more on Expedia's reviews because the refund information was clearer and easier to trust from the beginning.

- **Cognitive Gap:**

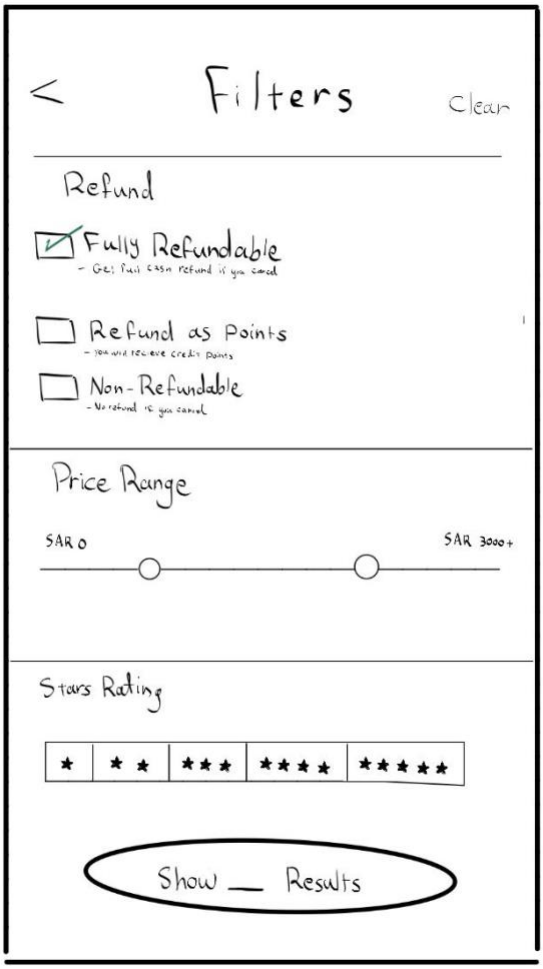
On AlMosafer, the cognitive gap was wider because she had to actively search for refund information, increasing anxiety and decision time. Expedia reduced this gap by placing critical information (refund status) within the primary visual area beside the hotel name.

Phase 3:

Redesigning the interface of the application



Sketch 1



Sketch 2

Sketch 1:

- Refundability information labels added under each hotel.
- Allows users to determine if the reservation is refundable before viewing hotels page.
- Saved time of navigating through different pages.
- Improved discoverability.
- Reduced decision making time.

Sketch 2:

- We changed the ambiguous filter "Free Cancellation" filter to a clear refund optons:
  - Full Refundable
  - Refund as Points
  - Non-Refundable
- Users get to understand refund policy quickly.
- Enhanced system status visibility.
- Less confusion.

These changes were added to tackle the usability problems discovered in Phase 2 with displaying refundability of bookings, as well as increasing the ease of completing booking task.